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Tutorial of Environmental Engineering and Science

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01

Ideal gas law

1. Ideal gas law

Ideal gas: All the gasses (e.g., O_2, CO_2, and air) in this course.

The **core expression** of the Ideal Gas Law is:

$$PV = nRT$$

Physical Quantity	Symbol	Core Meaning	SI Unit	Commonly Used Alternative Units	Conversion Relationship
Absolute Pressure	P	Absolute pressure of gas (not relative pressure)	Pa (Pascal)	atm (Atmosphere)	**1 atm=1.01×10^5 Pa**
Volume	V	Volume occupied by gas	m³ (Cubic meter)	L (Liter)	**1 m^3=1000 L**
Amount of Substance	n	Molar amount of gas	mol (Mole)	-	-
Ideal Gas Constant	R	Universal constant	8.31 J/(K·mol)	**R= 0.082056 L·atm·K^{-1}·mol^{-1}**	Use with matching P/V units.
Absolute Temperature	T	Thermodynamic temperature of gas	K (Kelvin)	°C (Celsius)	**T(K)=t(°C)+273.15**

Note: $\boldsymbol{J = N \cdot m = {}^{N}/_{m^2} \cdot m^3 = Pa \cdot m^3}$

1. Ideal gas law

Standard temperature and pressure (STP)

- ✓ 1 atm (1.01×10^5 Pa)
- ✓ 0℃ (273.15 K)

$PV = nRT$

Room Temperature Condition

- ✓ 1 atm (1.01×10^5 Pa)
- ✓ (25℃, $\mathbf{298.15}$ K)

$$V = \frac{1\ \text{mol}\times0.082056\ \text{L}\cdot\text{atm}\cdot\text{K}^{-1}\cdot\text{mol}^{-1}\times273.15\ \text{K}}{1\ \text{atm}} = \mathbf{22.414\ L}$$

$$V = \frac{1\ \text{mol}\times0.082056\ \text{L}\cdot\text{atm}\cdot\text{K}^{-1}\cdot\text{mol}^{-1}\times298.15\ \text{K}}{1\ \text{atm}} = \mathbf{24.465\ L}$$

22.414 L/mol (STP)

Molar volume of gas

24.465 L/mol (1 atm and 25℃)

Key conclusions: **1 mole** of ideal gas occupies **~22.414 L at STP** and **~24.465 L at 1 atm/25℃**.

ppmv ⟷ **Mas concentration**

ppmv = parts per million by volume

$$\mathbf{1}\ \text{ppmv}\ (CO_2) = \frac{\mathbf{1}\ \text{volume of gas}\ (CO_2)}{\mathbf{10^6}\ \text{volumes of air}}$$

Molar volume of gas, $V_m = \frac{nRT}{P}$.

$$\boldsymbol{mass\ concentration} = \frac{\text{ppmv}\times\boldsymbol{M_{Molar}}}{\boldsymbol{V_m}}$$

1. Ideal gas law

Example

A factory emits waste gas containing SO_2 at a flow rate of 1000 m^3/h (1 atm, 20°C). The SO_2 concentration is 500 ppmv. Calculate the **mass of SO_2 emitted per hour (kg)** (Molar mass of SO_2, $M = 64$ g/mol; $T = 20 + 273.15 = 293.15$ K).

Step 1: Calculate the molar volume of gas at 1 atm, 20°C

Not STP

$$PV = nRT \Rightarrow V_m = \frac{nRT}{P} = \frac{1 \times 0.082056 \times 293.15}{1} \approx 24.05 \text{ L/mol}$$

Step 2: Convert ppmv to mass concentration

Unit conversion!!

For 1 m^3 air

$$\rho_{so_2}(\text{mg/m}^3) = \frac{\text{ppmv} \times M}{V_m} = \frac{500 \times 10^{-6} \text{m}^3 \times 64 \text{g/mol}}{24.05 \text{ L/mol}} = \frac{500 \times 10^{-6} \text{m}^3 \times 64 \text{g/mol}}{24.05 \times 1000 \text{ m}^3/\text{mol}}$$

$$= 1.33056 \text{ g/m}^3 = 1.33056 \times 10^{-3} \text{ kg/m}^3$$

Step 3: Calculate the hourly emission mass of SO_2

Mass = Concentration × Gas Flow Rate

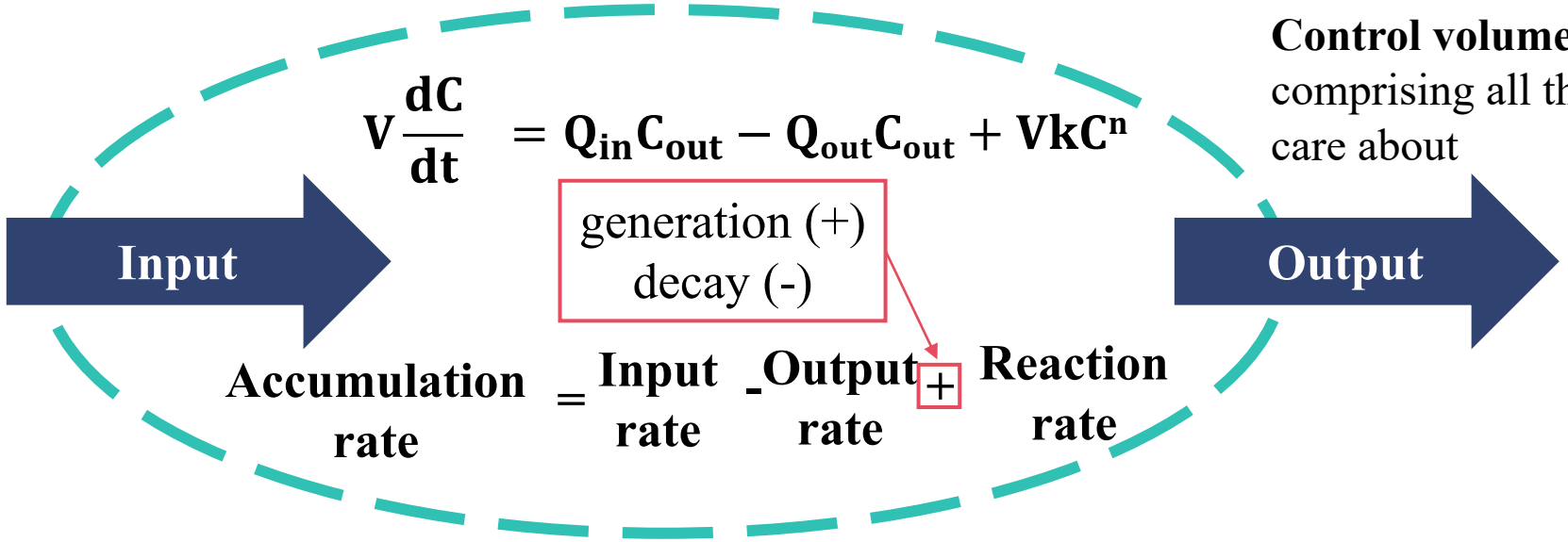
$$m = 1.33056 \times 10^{-3} \text{ kg/m}^3 \times 1000 \text{ m}^3/\text{h} \approx 1.33 \text{ kg/h}$$

Conclusion: The factory emits approximately **1.33 kg of SO_2 per hour**.

02

Mass balance

2. Mass balance



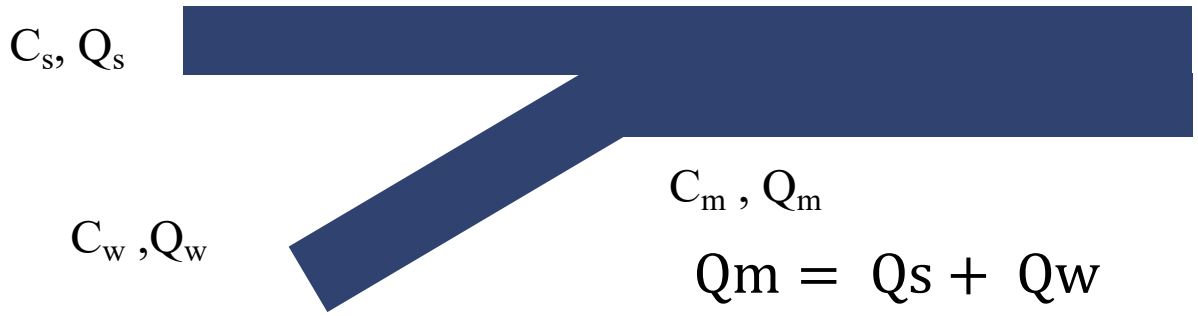
Control volume: a container comprising all the factors we care about

Steady state	accumulation rate = 0	$0 = Q_{in}C_{in} - Q_{out}C_{out} + VkC^n$
Conserved substance	reaction rate = 0	$V\frac{dC}{dt} = Q_{in}C_{in} - Q_{out}C_{out}$
Steady-state and conserved system	accumulation and reaction rate=0	$0 = Q_{in}C_{in} - Q_{out}C_{out}$
CSTR reactor	$C=C_{out}$	$0 = Q_{in}C_{in} - Q_{out}C_{out} + VkC^n$
PFR reactor (batch system)	Input rate = Output rate = 0	$V\frac{dC}{dt} = VkC^n$

2. Mass balance

- **Steady state conservative systems**

Accumulation rate = Input rate − Output rate + Reaction rate



$$Qm = Qs + Qw$$

$$\boldsymbol{QsCs + QwCw - QmCm = 0}$$

$$\boldsymbol{0 = Q_{in}C_{in} - Q_{out}C_{out}}$$

$$QsCs + QwCw - QmCm = 0$$

$$\rightarrow \boldsymbol{Cm = (QsCs + QwCw)/Qm}$$

$$= (\boldsymbol{QsCs + QwCw})/(\mathbf{Qs + Qw})$$

- **Batch systems with nonconservative pollutants**

Accumulation rate = ~~Input rate- Output rate~~ + Reaction rate

e.g., water evaporation from a cup

Accumulation rate = Reaction rate

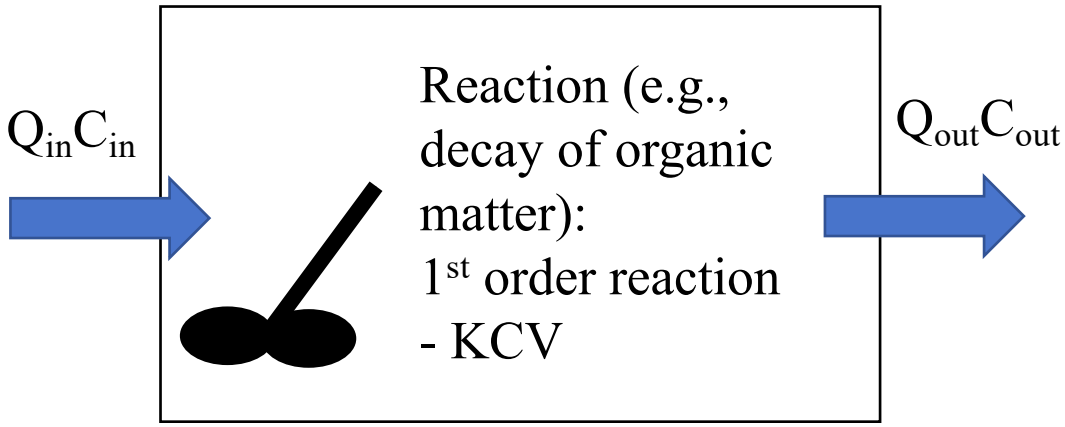
$$\frac{dC}{dt} = \text{reaction rate}$$

- **k (zero-order) →** $\mathbf{C = C_0 + kt}$
- **kC (first-order) →** $\mathbf{C = C_0\, e^{kt}}$
- kC^2 (second-order)

e.g., bacteria growth, population dynamics

2. Mass balance

- **Continuously stirred tank/Complete-mix reactor (CSTR)**



$$Q_{\mathrm{in}}\mathrm{C}_{\mathrm{in}} - \mathrm{Q}_{\mathrm{out}}\mathrm{C}_{\mathrm{out}} - kC_{\mathrm{out}}V = 0$$

$$Cout = Q\mathrm{C}_{\mathrm{in}}/(Q + kV) = \mathrm{C}_{\mathrm{in}}/(1 + kV/Q)$$

Assume 1st Order Reaction

$$\boldsymbol{C_{out} = \frac{C_{in}}{1 + k\theta}}$$

Where θ is the hydraulic retention time (HRT), θ = V/Q

Typical applications: **Long pipes, rivers, disinfection contact tanks, packed-bed reactors**.

- **Plug flow reactor (PFR)**



Assume 0th Order Reaction

$$\frac{dC}{dt} = -k, C_{out} - C_{in} = -k\theta \qquad \boldsymbol{C_{out} = C_{in} - k\theta}$$

Assume 1st Order Reaction

$$\frac{dC}{dt} = -kC, \int_{C_{in}}^{C_{out}} \frac{dC}{C} = -k\int_0^{\theta} d\,t \qquad \ln\left(\frac{C_{out}}{C_{in}}\right) = -k\theta$$

$$\boldsymbol{C_{out} = C_{in}e^{-k\theta}}$$

Typical applications: **Lakes, well-mixed ponds, small treatment tanks with mechanical mixers.**

2. Mass balance

Question

Wastewater ($C_{in} = 500$ mg/L, $Q = 100$ L/h) is treated by a reactor with volume $V = 500$ L and first-order decay ($k = 0.5\ \mathrm{h}^{-1}$). Calculate the effluent concentration C_{out} for:

(1) CSTR; (2) PFR. Compare the two results and explain the difference.

Step 1: Calculate HRT

$$\theta = V/Q = \frac{500}{100} = 5\ \mathrm{h}$$

$$\boldsymbol{C_{out} = \frac{QC_{in}}{Q + kv} = \frac{C_{in}}{1 + k\theta}}$$

(1) CSTR: $C_{out} = \frac{C_{in}}{1+k\theta} = \frac{500}{1+(0.5\times5)} = \frac{500}{3.5} \approx 142.86$ mg/L

$$\boldsymbol{C_{out} = C_{in} - k\theta}$$

(2) PFR: $C_{out} = C_{in}e^{-k\theta} = 500\times e^{-(0.5\times5)} = 41.05$ mg/L

Step 2: Comparison

PFR's C_{out} (41.05 mg/L) is **much lower** than CSTR's (142.86 mg/L) → PFR has a **higher treatment efficiency**.

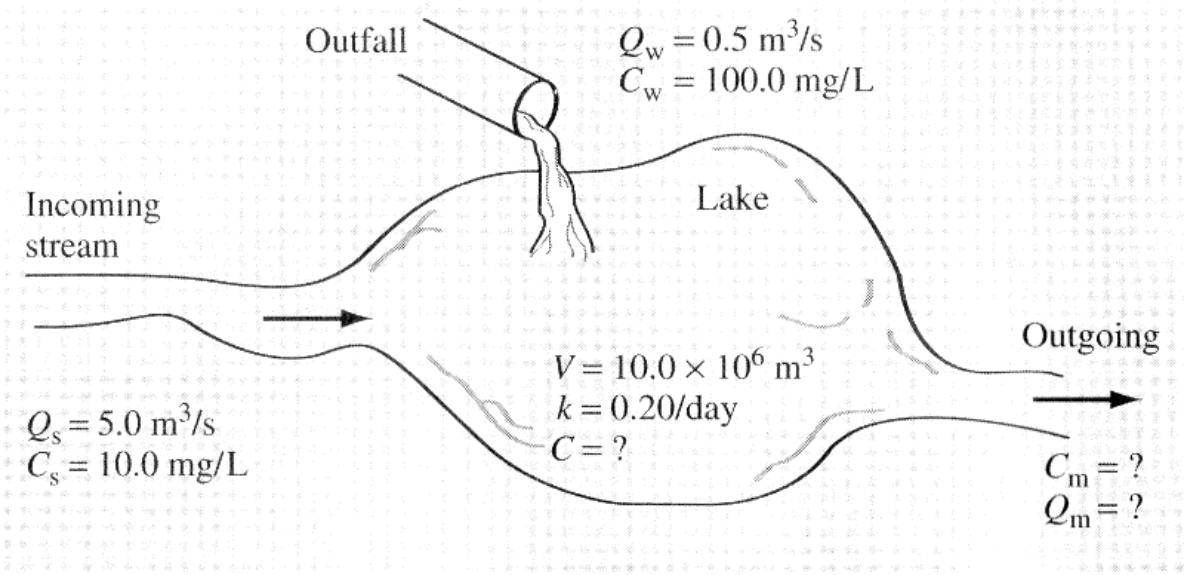
2. Mass balance

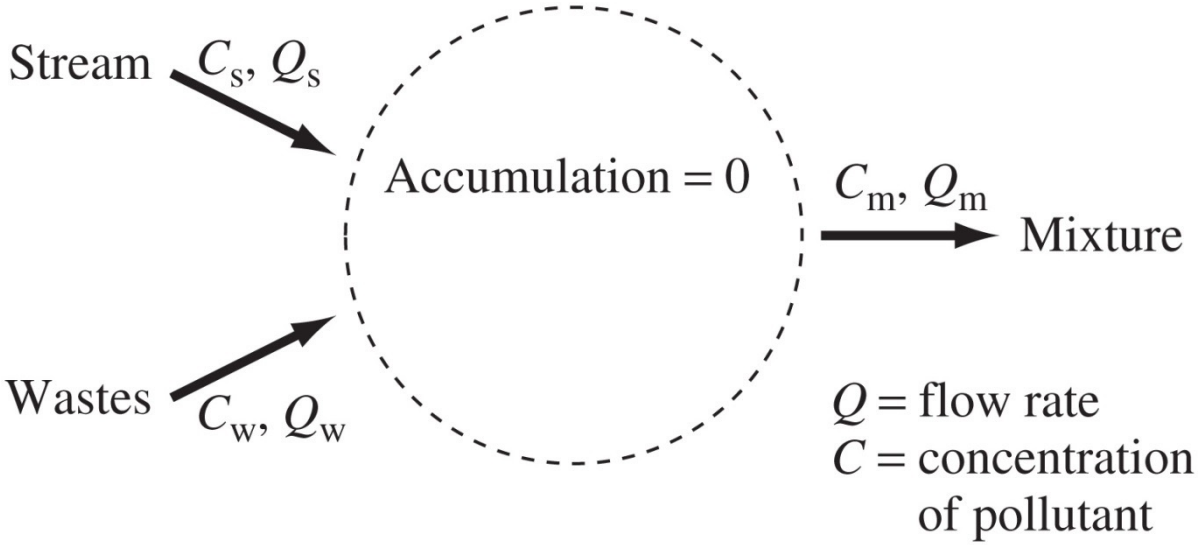
Question

A lake with a constant volume of 10×10^6 m^3 receives flow from a stream and a wastewater outfall.

- Stream flow: $Q_s = 5\ m^3/s$, with pollutant concentration $C_s = 10$ mg/L.
- Wastewater flow: $Q_w = 0.5\ m^3/s$, with pollutant concentration $C_w = 100$ mg/L.
- The pollutant decays with a first-order reaction rate coefficient $k = 0.2\ day^{-1}$.
- The lake is **completely mixed (CSTR)** and is at steady state.

Find the steady-state concentration of the pollutant leaving the lake (C_m).





2. Mass balance

Step 1: Convert the reaction rate to consistent units (per second)

Since flow rates are in m^3/s, we must convert k to s^{-1}:

$$k = \frac{0.2\text{day}-1}{24\text{h}\times 3600\text{s}} = 2.315\times 10^{-6}\ s^{-1}$$

Step 2: Calculate the total outflow

For a CSTR at steady state, inflow equals outflow:

$$Q_m = Q_s + Q_w$$
$$= 5.0 m^3/s + 0.5 m^3/s = 5.5\ m^3/s$$

Step 3: Write the mass balance equation

At steady state (Accumulation = 0):

$$0 = inputrate1 + Inputrate2 - Outputrate - reactionrate(decay)$$

$$Q_s C_s + Q_w C_w - Q_m C_m - kVC_m = 0$$

Factor out C_m on the right side:

$$Q_s C_s + Q_w C_w = C_m(Q_m + kV)$$

2. Mass balance

Step 4: Calculate the decay term (kV)

$$kV = (2.315\times10^{-6}\ \mathrm{s}^{-1})\times(10\times10^{6}\ \mathrm{m}^{3}) = 23.15\ \mathrm{m}^{3}/\mathrm{s}$$

Step 5: Substitute values and solve for C_m

$$(5\times10) + (0.5\times100) = C_m(5.5 + 23.15)$$
$$50 + 50 = C_m(28.65)$$
$$100 = 28.65\times C_m$$
$$C_m = \frac{100}{28.65} = 3.49\ \mathrm{g/m}^{3}$$

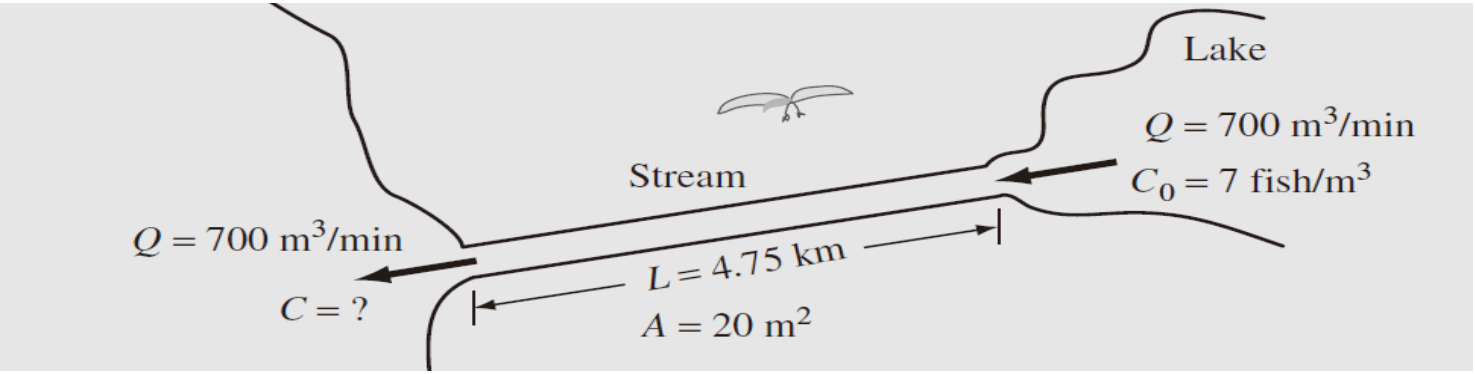
Since $1\ \mathrm{mg/L} = 1\ \mathrm{g/m}^{3}$:

$$C_m \approx 3.5\ \mathrm{mg/L}$$

2. Mass balance

Question

Every year, herons, seagulls, eagles, and other birds gather along a 4.75-km stretch of a stream connecting a lake to the ocean. They feed on fingerling salmon migrating downstream. The birds **consume 10,000 fingerlings per kilometer of stream per hour**, regardless of the number of salmon in the water (**this is a zero-order reaction** – limited only by the birds' eating speed). The stream has an average cross-sectional area of 20 m², and the salmon move downstream with the stream flow rate of 700 m³/min. If the water entering the stream contains 7 fingerlings per m³, what is **the concentration of salmon** that survive and reach the ocean while the birds are feeding?



Given Data

Stream length, $L = 4.75$ km $= 4750$ m

Cross-sectional area, $A = 20$ m^2

Flow rate, $Q = 700$ m^3/min

Initial concentration, $C_0 = 7$ fish/m^3

Bird consumption rate = 10,000 fish/(km · hour)

The stream is best modelled as a **Plug Flow Reactor (PFR)** because the **water flows downstream without mixing backwards**. In a PFR, the reaction occurs as the water travels along the length, so we treat it like a batch reactor moving downstream.

2. Mass balance

Step 1: Calculate the total volume of the stream stretch

$$V = L \times A = 4750\ \text{m} \times 20\ \text{m}^2 = 95{,}000\ \text{m}^3$$

Step 2: Calculate the residence time (t) – the time water (and salmon) spend in the stream

Convert flow to m^3/hour:

$$Q = 700\ \text{m}^3/\text{min} \times 60\ \text{min/hr} = 42{,}000\ \text{m}^3/\text{hr}$$

$$t = \frac{V}{Q} = \frac{95{,}000}{42{,}000} = 2.26\ \text{hours}$$

Step 3: Convert the bird consumption rate to suitable units (fish per m³ per hour)

The birds eat 10,000 fish per km per hour. The water volume in 1 km of stream is:

$$V_{1\text{km}} = 1000\ \text{m} \times 20\ \text{m}^2 = 20{,}000\ \text{m}^3$$

So the zero-order decay rate constant k is:

$$k = \frac{10{,}000\ \text{fish/hr}}{20{,}000\ \text{m}^3} = 0.5\ \text{fish}/(\text{m}^3 \cdot \text{hr})$$

2. Mass balance

Step 4: Apply the zero-order decay equation for a PFR
For zero-order decay:

$$C = C_0 - k \cdot t$$

Substitute the values:

$$C = 7 \text{ fish/m}^3 - (0.5 \text{ fish/(m}^3 \cdot \text{hr)} \times 2.26 \text{ hr})$$
$$C = 7 - 1.13 = 5.87 \text{ fish/m}^3$$

Therefore, aout 5.9 out of every 7 fingerlings per cubic metre survive the bird feeding and reach the ocean.

2. Mass balance

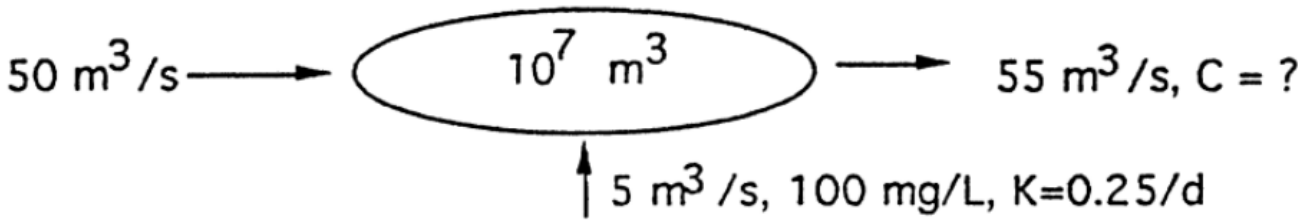
Question

A lake with a constant volume of 10×10^6 m^3 is fed by a pollution-free stream with a flow rate of 50 m^3/s. A factory dumps 5 m^3/s of a nonconservative waste with a concentration of 100 mg/L into the lake. The pollutant has a reaction rate coefficient K of 0.25 day^{-1}. Assuming the pollutant is **well mixed in the lake**, find the **steady-state concentration** of the pollutant in the lake.

Given Data:

Lake volume, $V = 10\times10^6$ m^3,Clean stream flow, $Q_s = 50$ m^3/s,Waste flow, $Q_w = 5$ m^3/s

Waste concentration, $C_w = 100$ mg/L, Reaction rate, $K = 0.25$ day^{-1}, Total outflow, $Q_{out} = Q_s + Q_w = 55$ m^3/s



$$\text{Input} = \text{Output} + \text{Decay}\ ,\ \text{where decay} = \text{KCV}$$

Step 1: Convert the reaction rate to consistent units (per second)

$$K = \frac{0.25\text{day}^{-1}}{24\text{h}\times3600\text{s}} = 2.89\times10^{-6}\ \text{s}^{-1}$$

2. Mass balance

Step 2: Write the steady-state mass balance equation

At steady state:

$$\text{Input} - \text{Output} - \text{Decay}=0$$

$$Q_w C_w - Q_{out}\, C - KCV = 0$$

$$Q_w C_w = Q_{out} C + KCV$$

Step 3: Substitute the numerical values

$$5\times100 = 55C + (2.89\times10^{-6})\times C\times(10\times10^{6})$$

$$500 = 55C + 28.9C$$

$$500 = 83.9C$$

Step 4: Solve for C

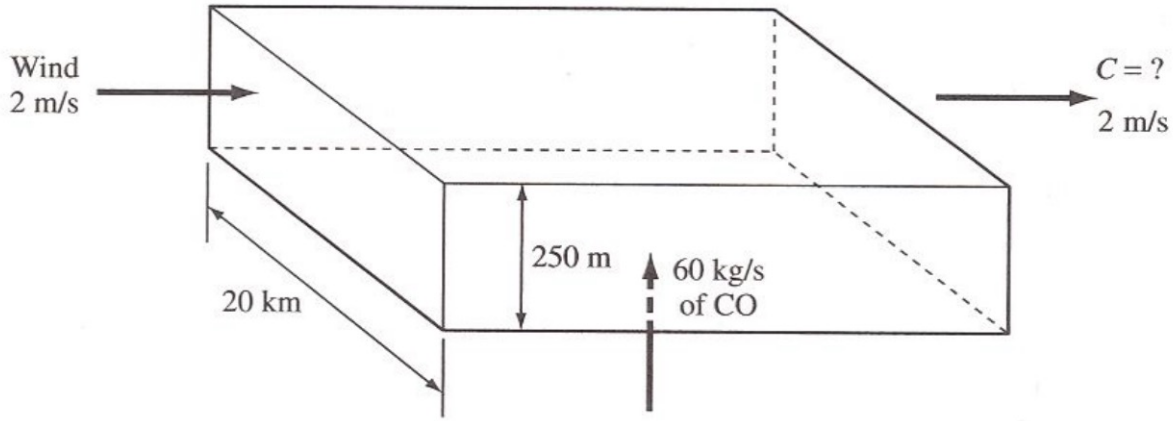
$$C = \frac{500}{83.9} = 5.96 \text{ mg/L}$$

The steady-state concentration of the pollutant is 5.96 mg/L in the lake.

2. Mass balance

Question:

A simple way to model air pollution over a city is with a box model that assumes complete mixing. The town has an inversion layer at 250 m, a 20-km horizontal distance perpendicular to the wind, a windspeed of 2 m/s, and a carbon monoxide (CO) emission rate of 60 kg/s. **Assume CO is conservative and completely mixed in the box**. What would be the **CO concentration** in the box?



Given Data:

Inversion height, $H = 250$ m

Horizontal width (perpendicular to wind), $W = 20$ km $= 20{,}000$ m

Wind speed, $U = 2$ m/s

CO emission rate, $M = 60$ kg/s $= 60\times10^6$ mg/s

Step 1: Calculate the volumetric flow rate of air through the box

$$Q = W\times H\times U = 20{,}000\times250\times2 = 10{,}000{,}000\ \text{m}^3/\text{s}$$

Step 2: Apply the steady-state mass balance for a conservative pollutant

At steady state (no reaction, accumulation = 0):

$$\text{Input rate} = \text{Output rate}$$
$$M = Q\times C$$

Step 3: Solve for the concentration C

$$C = \frac{M}{Q} = \frac{60\times10^6}{20{,}000\times250\times2} = \frac{60{,}000{,}000}{10{,}000{,}000} = 6\ \text{mg/m}^3$$

CO concentration is 6 mg/m^3 in the box